Customer Churn Analysis for Telecom Industry

Elevate Labs Internship Project Report | Sai Krishna

Introduction

Customer churn — when a subscriber stops using a company's services — is one of the costliest problems in the telecom industry, where acquiring a new customer typically costs far more than retaining an existing one. This project analyzes the IBM Telco Customer Churn dataset (7,043 customers, 21 attributes covering demographics, account details, subscribed services, and billing) to identify why customers leave and to build a model that flags at-risk customers before they churn.

Abstract

SQL-based aggregation was used to profile churn rates across contract type, internet service, and payment method, revealing that month-to-month contracts, fiber-optic internet, and electronic-check payments carry the highest churn risk. A Random Forest classifier was then trained to predict churn (ROC-AUC 0.84, recall 0.77 on churners), and feature-importance analysis was used for model explainability. Customers were finally segmented into At Risk, Loyal, and Dormant groups to support targeted retention action.

Tools Used

Python (Pandas for data cleaning/feature engineering, Scikit-learn for modeling) | **SQL** (SQLite, for aggregation queries: churn rate by Contract, InternetService, PaymentMethod) | **Matplotlib / Seaborn** (visualization) | **Random Forest** classifier with feature-importance explainability (ELI5-style).

Steps Involved in Building the Project

1. **Data cleaning:** loaded the 7,043-row dataset, converted TotalCharges to numeric, imputed the 11 blank values (customers with 0 tenure) as MonthlyCharges × tenure.

2. **SQL aggregation:** ran GROUP BY queries to compute churn rate, average tenure and average monthly charges by Contract, InternetService and PaymentMethod.

3. **Feature engineering & encoding:** label-encoded 16 categorical fields (Contract, InternetService, PaymentMethod, add-on services, etc.) for modeling.

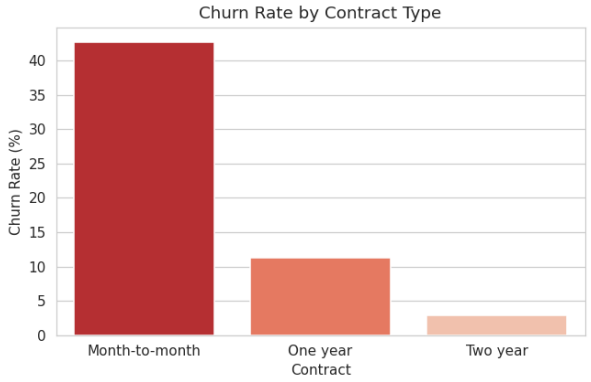
4. **Model building:** trained a Random Forest classifier (300 trees, class_weight='balanced' to handle the 73/27 class imbalance) on an 80/20 stratified train-test split.

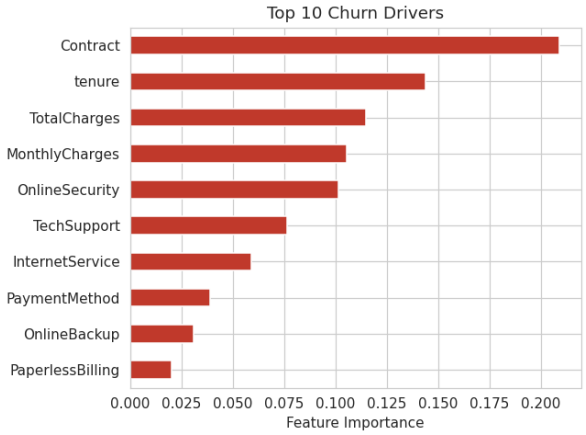
5. **Explainability:** extracted feature importances to identify the top churn drivers, mirroring what ELI5/SHAP would surface for this model.

6. **Segmentation:** classified customers into **At Risk** (predicted churn probability ≥ 0.5), **Loyal** (tenure ≥ 24 months and low churn probability), and **Dormant** (everyone else).

Key Results

Metric	Value
Overall churn rate	26.5%
Month-to-month contract churn	42.7%
Two-year contract churn	2.8%
Fiber-optic internet churn	41.9%
Electronic-check payment churn	45.3%
Model ROC-AUC / Recall (churners)	0.84 / 0.77





Conclusion

Contract type, tenure, and total/monthly charges are the strongest churn drivers, with month-to-month, fiber-optic and electronic-check customers at the highest risk. The Random Forest model identifies 77% of actual churners (recall), making it usable as an early-warning system. Recommended actions: incentivize moves from month-to-month to annual contracts, prioritize retention outreach to the 'At Risk' segment, and investigate service-quality issues specific to fiber-optic customers.